

Κανόνες παραγώγισης

- $f(x) = A \rightarrow f'(x) = 0$
- $f(x) = x^\alpha \rightarrow f'(x) = \alpha x^{\alpha-1}$
- $f(x) = A + g(x) \rightarrow f'(x) = g'(x)$
- $f(x) = Ag(x) \rightarrow f'(x) = Ag'(x)$

Εάν οι συναρτήσεις $f(x)$ και $g(x)$ έχουν παράγωγο, τότε

- $[f(x) + g(x)]' = f'(x) + g'(x)$
- $[f(x) - g(x)]' = f'(x) - g'(x)$
- $[f(x)g(x)]' = f'(x)g(x) + f(x)g'(x)$
- $\left[\frac{f(x)}{g(x)}\right]' = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}, g(x) \neq 0$
- $f(x) = a^x \rightarrow f'(x) = a^x \ln(a), a > 0$
- $f(x) = e^x \rightarrow f'(x) = e^x (e \approx 2,71828)$
- $f(x) = \ln(x) \rightarrow f'(x) = \frac{1}{x}$
- $f(x) = \log_a(x) \rightarrow f'(x) = \frac{1}{x \ln(a)}, a > 0 \text{ και } a \neq 1$